Foundations of Federated learning

[Federated learning]

1.0 Content Block

Personalized federated learning by pruning (Sub-FedAvg) Federated learning methods have poor global performance under non-IID settings. This motivates clients to yield personalized models in federation. To change this, Vahidian et al. recently introduced the algorithm Sub-FedAvg which does hybrid pruning (structured and unstructured pruning) with averaging on the intersection of clients' drawn subnetworks. This simultaneously addresses communication efficiency, resource constraints and personalized models accuracies. Sub-FedAvg also extends the "lottery ticket hypothesis" of centrally trained neural networks to federated learning, with the research question: "Do winning tickets exist for clients' neural networks being trained in federated learning? If so, how to effectively draw the personalized subnetworks for each client?" Sub-FedAvg experimentally answers "yes" and proposes two algorithms to effectively draw the personalized subnetworks.

2.0 Content Block

Personalized federated learning In many application domains, not all clients are the same. Clients may share the same model architecture, but each have a different data distribution—which is known as Personalized federated learning (PFL), or even have different model architectures, like mobile phones and IoT devices—which is known as Heterogeneous federated learning. The key challenge in training a PFL model is to keep each client different from others, and at the same time benefit from generalizing across clients.

3.0 Content Block

Heterogeneous federated learning Most of the existing federated learning strategies assume that local models share the same global model architecture. Recently, a new federated learning framework named HeteroFL was developed to address heterogeneous clients equipped with very different computation and communication capabilities. The HeteroFL technique can enable the training of heterogeneous local models with dynamically varying computation and non-IID data complexities while still producing a single accurate global inference model.

4.0 Content Block

where K {\displaystyle K} is the number of nodes, x i {\displaystyle \mathbf {x} _{i}} are the weights of model as viewed by node i {\displaystyle i} , and f i {\displaystyle f_{i}} is node i {\displaystyle i} 's local objective function, which describes how model weights x i {\displaystyle \mathbf {x} _{i}} conforms to node i {\displaystyle i} 's local dataset. The goal of federated learning is to train a common model on all of the nodes' local datasets, in other words:

5.0 Content Block

Federated learning (also known as collaborative learning) is a machine learning technique in a setting where multiple entities (often called clients) collaboratively train a model while keeping their data decentralized, rather than centrally stored. A defining characteristic of federated learning is data heterogeneity. Because client data is decentralized, data samples held by each client may not be independently and identically distributed. Federated learning is generally concerned with and motivated by issues such as data privacy, data minimization, and data access rights. Its applications involve a variety of research areas including defence, telecommunications, the Internet of things, and pharmaceuticals.

6.0 Content Block

Federated learning parameters Once the topology of the node network is chosen, one can control different parameters of the federated learning process (in addition to the machine learning model's own hyperparameters) to optimize learning:

7.0 Content Block

Use cases Federated learning typically applies when individual actors need to train models on larger datasets than their own, but cannot afford to share the data in itself with others (e.g., for legal, strategic or economic reasons). The technology yet requires good connections between local servers and minimum computational power for each node.

8.0 Content Block

Current research topics Federated learning has started to emerge as an important research topic in 2015 and 2016, with the first publications on federated averaging in telecommunication settings. Before that, in a thesis work titled "A Framework for Multi-source Prefetching Through Adaptive Weight", an approach to aggregate predictions from multiple models trained at three location of a request response cycle with was proposed. Another important aspect of active research is the reduction of the communication burden during the federated learning process. In 2017 and 2018, publications have emphasized the development of resource allocation strategies, especially to reduce communication requirements between nodes with gossip algorithms as well as on the characterization of the robustness to differential privacy attacks. Other research activities focus on the reduction of the bandwidth during training through sparsification and quantization methods, where the machine learning models are sparsified and/or compressed before they are shared with other nodes. Developing ultra-light DNN architectures is essential for device-/edge-learning and recent work recognises both the energy efficiency requirements for future federated learning and the need to compress deep learning, especially during learning. Recent research advancements are starting to consider real-world propagating channels as in previous implementations ideal channels were assumed. Another active direction of research is to develop Federated learning for training heterogeneous local models with varying computation complexities and producing a single powerful global inference model. A learning framework named Assisted learning was recently developed to improve each agent's learning capabilities without transmitting private data, models, and even learning objectives. Compared with Federated learning that often requires a central controller to orchestrate the learning and optimization, Assisted learning aims to provide protocols for the agents to optimize and learn among themselves without a global model.

9.0 Content Block

Iterative learning The iterative process of federated learning is composed of a series of fundamental client-server interactions, each of which is known as a federated learning round. Each round of this process consists in transmitting the current global model state to participating nodes, training local models on these local nodes to produce a set of potential model updates at each node, and then aggregating and processing these local updates into a single global update and applying it to the global model. In the methodology below, a central server is used for aggregation, while local nodes perform local training depending on the central server's orders. However, other strategies lead to the same results without central servers, in a peer-to-peer approach, using gossip or consensus methodologies. Assuming a federated round composed by one iteration of the learning process, the learning procedure can be summarized as follows:

10.0 Content Block

Covariate shift: local nodes may store examples that have different statistical distributions compared to other nodes. An example occurs in natural language processing datasets where people typically write the same digits/letters with different stroke widths or slants. Prior probability shift: local nodes may store labels that have different statistical distributions compared to other nodes. This can happen if datasets are regional and/or demographically partitioned. For example, datasets containing images of animals vary significantly from country to country. Concept drift (same label, different features): local nodes may share the same labels but some of them correspond to different features at different local nodes. For example, images that depict a particular object can vary according to the weather condition in which they were captured. Concept shift (same features, different labels): local nodes may share the same features but some of them correspond to different labels at different local nodes. For example, in natural language processing, the sentiment analysis may yield different sentiments even if the same text is observed. Unbalanced: the amount of data available at the local nodes may vary significantly in size. The loss in accuracy due to non-iid data can be bounded through using more sophisticated means of doing data normalization, rather than batch normalization.